

Broadening Our View: Assistive Technology for Cerebral Visual Impairment

Bhanuka Gamage
bhanuka.gamage@monash.edu
Monash University
Melbourne, Australia

Thanh-Toan Do
toan.do@monash.edu
Monash University
Melbourne, Australia

Leona Holloway
leona.holloway@monash.edu
Monash University
Melbourne, Australia

Nicholas Price
nicholas.price@monash.edu
Monash University
Melbourne, Australia

Nicola McDowell
n.mcdowell@massey.ac.nz
Massey University
Auckland, New Zealand

Arthur Lowery
arthur.lowery@monash.edu
Monash University
Melbourne, Australia

Kim Marriott
kim.marriott@monash.edu
Monash University
Melbourne, Australia

ABSTRACT

Over the past decade, considerable research has been directed towards assistive technologies to support people with vision impairments using machine learning, computer vision, image enhancement, and/or augmented/virtual reality. However, this has almost totally overlooked a growing demographic: people with Cerebral Visual Impairment (CVI). Unlike Ocular Vision Impairments (OVI), CVI arises from damage to the brain's visual processing centres. This paper introduces CVI and reveals a wide research gap in addressing the needs of this demographic. Through a scoping review, we identified 14 papers at the intersection of these technologies and CVI. Of these, only three papers described assistive technologies focused on people living with CVI, with the others focusing on diagnosis, understanding, simulation or rehabilitation. Our findings highlight the opportunity for the Human-Computer Interaction and Assistive Technologies research community to explore and address this underrepresented domain, thereby enhancing the quality of life for people with CVI.

CCS CONCEPTS

- **Human-centered computing** → **Accessibility technologies**;
- **General and reference** → **Surveys and overviews**.

KEYWORDS

cerebral visual impairment, assistive devices, computer vision, machine learning, augmented reality, virtual reality

ACM Reference Format:

Bhanuka Gamage, Leona Holloway, Nicola McDowell, Thanh-Toan Do, Nicholas Price, Arthur Lowery, and Kim Marriott. 2024. Broadening Our

View: Assistive Technology for Cerebral Visual Impairment. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3613905.3650740>

1 INTRODUCTION

Vision is not solely a product of the eyes; it also relies on intricate processes of the brain. **Cerebral Visual Impairment (CVI)** is a visual dysfunction that is distinct from Ocular Vision Impairment (OVI), and caused by injury or disruption to the brain's visual processing centres [79]. CVI has emerged as the predominant cause of childhood vision impairment in developed nations [34, 61, 80], affecting around 30-40% of visually impaired children [69]. As these children transition into adulthood, CVI is poised to become the leading cause of vision impairment [19].

Recent breakthroughs in deep learning, particularly in computer vision and multimodal large language models [70, 90, 95, 97], have sparked a surge of interest in enhancing assistive technologies for people with vision impairments. Concurrently, a renewed interest in augmented and virtual reality has led to the development of improved passthrough capabilities on most modern devices [60, 65, 68, 76, 94]. We define **Vision-Based Assistive Technologies (VBAT)** as devices incorporating machine learning, computer vision, image enhancement, and/or augmented/virtual reality to enhance the quality of life for people with vision impairments. These can perform object detection, image enhancement and scene simplification and are built into smart glasses, smart phones or other wearables. This study aims to address the primary question: **What is the current state of research at the intersection of CVI and VBAT, and what opportunities exist in this domain?** The main contributions of our study are:

- **Scoping Review:** We conduct a scoping review [67] that provides insight into the current research landscape concerning CVI and technologies for VBAT. It reveals a paucity of research and a predominant focus on understanding CVI rather than assistance.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650740>

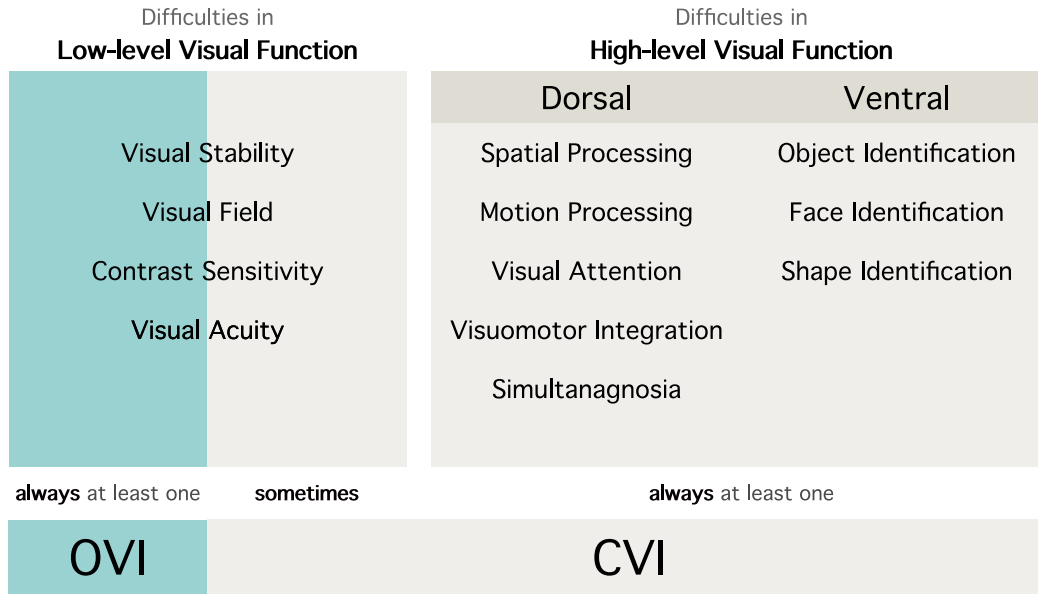


Figure 1: Overview of low-level and high-level visual difficulties for people with CVI. Note that this list is not exhaustive.

- **Opportunities of VBAT:** We identify potential applications of VBAT that will significantly enhance the quality of life for people with CVI and propose promising areas for future work.
- **Raising Awareness of CVI in HCI and Assistive Technology (AT) Communities:** We discuss the importance and timing for the HCI and AT communities to address these opportunities.

We hope that our paper will provide guidance and inspiration to researchers in the fields of HCI and AT to address the unmet needs of the CVI community.

2 CEREBRAL VISUAL IMPAIRMENT (CVI)

Cerebral Visual Impairment (CVI) is the result of damage to the brain's visual processing centres, rather than any physical damage to the eye [56, 77]. CVI is primarily observed in individuals with neurological conditions like cerebral palsy, stroke, or traumatic brain injury. People with CVI face issues with interpreting and processing visual information, including difficulties in visual recognition, perception, understanding of their visual environment, and maintaining visual attention.

The terminology and definitions related to this condition, including Cerebral Visual Impairment [56, 74], Cortical Vision Impairment [31, 77, 92], and Neurological Vision Impairment [88], vary subtly and are a subject of ongoing debate [59, 62]. We will use Cerebral Visual Impairment (CVI) to encompass a range of visual processing challenges resulting from damage to the brain's visual processing centres.

Ocular Visual Impairment (OVI) refers to vision impairments arising from problems within the eye or its associated structures,

such as the retina, optic nerve, or cornea [59]. OVI examples include conditions like cataracts, glaucoma, and macular degeneration, which can cause a range of vision impairments from partial (low-vision) to total blindness.

2.1 CVI Diagnosis and Prevalence

In the last decade, our understanding of CVI has significantly expanded [56, 77]. Consequently, there has been an upsurge in the diagnosis of CVI among children. This increased recognition can be attributed to the dissemination of knowledge regarding its primary causes, especially premature birth, among other factors [14, 26, 30, 44, 87]. CVI is frequently under-diagnosed, primarily because other forms of brain damage often coexist, causing a range of physical and cognitive impairments. In some instances, CVI might be mistaken for conditions like autism, learning disabilities, or behavioural challenges, further complicating accurate diagnosis [86, 93].

Numerous studies have focused on identifying children with CVI. Williams et al. [93] screened a substantial cohort of children and found a CVI prevalence rate of 3.4% among those in mainstream classrooms. In addition to the large prevalence of CVI in children, it is believed that many adults grapple with undiagnosed CVI. Furthermore, adults can also develop CVI later in life, often stemming from factors such as strokes, traumatic brain injuries, and other neurological conditions [52, 98].

At present, assessments and classifications of visual capability focus on OVI and primarily rely on criteria related to visual acuity and visual field. People with CVI may have normal visual acuity but struggle with higher-level visual functions [39, 78]. For instance, a child with CVI may have difficulty recognising a parent only when in a crowded room, or finding a particular toy only when it's mixed with other objects [24]. This presents clinicians with challenges

Table 1: Similarities and Differences among people who are blind, have low vision, and have CVI

Criteria	Blindness	Low Vision	CVI
Interaction	Audio, Haptic	Audio, Haptic, Visual (residual vision)	Audio, Haptic, Visual (normal vision)
Visual Dysfunction	Low-Level	Low-Level	Low-Level & High-Level
Complexity Impact on Vision	None	No Impact	High Impact
Effects of Vision Rehabilitation	Little Impact	Little Impact	High Impact
Association with Other Neurological Conditions	Rarely	Rarely	Frequently
Visual Difficulties Examples	Minimal to no visual perception	Difficulty seeing objects, sensitive to contrast, identifying colour	Finding objects in clutter, crowding of text, route finding, finding a relative or friend in a crowded room

when diagnosing people with CVI because the existing framework does not sufficiently address their needs. It also means that many individuals with CVI do not meet the legal definition of blindness as per current standards [45]. This underscores the necessity for a broader understanding of CVI along with advocacy for a revision of the definitions of vision impairment to encompass people with CVI.

2.2 Low-Level Functional Vision vs. High-Level Functional Vision

People with CVI commonly experience difficulties in both low-level and high-level functional vision [16, 21]. Low-level functional vision encompasses fundamental aspects of visual perception like visual acuity, contrast sensitivity, visual field, and visual stability, similar to people with OVI.

Conversely, high-level functional vision pertains to how individuals interpret and respond to visual information, including processes like visual recognition, comprehension, visual attention, and interaction with the environment. Studies indicate that these high-level vision impairments are associated with damage in two critical visual processing pathways in the brain—the Dorsal Stream and Ventral Stream [14].

The two-stream model proposed by Goodale [32], consisting of the ‘dorsal stream’—connecting the occipital to the parietal cortex and the ‘ventral stream’—connecting the occipital to the inferior temporal cortex, offers a valuable framework for understanding higher-order visual processing difficulties in CVI [35, 66]. Dysfunction in the dorsal stream typically manifests as vision impairments related to spatial and motion processing, difficulties in visual attention, challenges in visuomotor integration, and simultanagnosia (the inability to perceive more than one object at a time) [14, 28]. Meanwhile, damage along the ventral visual processing stream is associated with difficulties in object identification [14, 32, 35], such as recognising faces and shapes [6, 38]. However, dorsal and ventral stream difficulties often co-occur, suggesting that damage cannot be localised to a single brain area [12, 25, 57]. Figure 1 provides an overview of the types of visual difficulties for people with CVI.

Recent advancements have provided tools to identify high-level visual function difficulties in people with CVI [21, 63]. These include

questionnaires for initial assessment [21], and an iPad app that induces visual crowding [62, 64].

2.3 Assistive Technologies for CVI

A large body of research within the field of HCI and AT has been dedicated to aiding people with OVI [7, 9, 22, 41, 43, 47, 53, 71, 73, 81, 89]. These studies can be categorised into two primary areas: studies focusing on audio or tactile solutions for people who are blind (Ocular Blindness) [4, 17, 42, 83, 91, 96], and studies focusing on visual devices for people with low vision (Ocular Low Vision) [49, 100–103].

Given the shared reliance on audio and haptic interactions in OVI and CVI, it is tempting to assume that research designed for OVI could be readily repurposed to benefit those with CVI. This is especially true for Ocular Low Vision studies [49, 102, 103], as they also encompass the visual modality as well. However, this does not account for the two levels of functional vision difficulties within the CVI population. In the case of people with low-level functional vision difficulties, we concur that devices and applications centered on OVI can indeed be applicable. However, for people with high-level functional vision difficulties, evidence from the educational field indicates that strategies for people with OVI may not be effective [59]. To elaborate, we have identified four main differences:

- **Vision-Centric Interaction:** People with CVI primarily rely on their vision, whereas people with low vision are more likely to prioritise audio and haptic interactions [101]. When combined simultaneously, multiple modalities, such as visual, audio, and haptic, can be overwhelming for people with CVI [48, 74].
- **Complexity Impact on Vision:** People with CVI experience escalating difficulties as visual complexity increases, while people with OVI maintain relatively stable performance [12]. Hence, the nature of device interaction becomes crucial, particularly in high-complexity tasks, as people with CVI require less taxing and overwhelming interactions compared to neurotypical and OVI participants.
- **Neuroplasticity and Brain Modification:** Unlike OVI, CVI is a brain-based disorder, enabling assistive devices to induce permanent changes in the brain’s visual pathways through

neuroplasticity [14]. This effect can be both advantageous, potentially improving vision over time, or detrimental, possibly leading to lasting vision changes due to reliance on assistive devices. Therefore, researchers studying assistive devices for CVI should exercise caution and consider long-term effects through extended studies.

- **Frequent Association with Other Neurological Conditions:** In contrast to OVI, CVI often co-occurs with other neurological conditions like cerebral palsy [56], which need to be considered within the research approach.

Table 1 provides a concise summary of the key similarities and distinctions among people who have ocular blindness, ocular low vision and CVI.

3 OPPORTUNITIES

Recent advancements in artificial intelligence, particularly in computer vision and multimodal large language models [70, 90, 95, 97], have sparked interest in developing assistive technologies for people with OVI [29, 51, 53]. These range from basic object recognition [7, 9, 41], to advanced autonomous guidance systems [22, 71] and are built into devices from smart-glasses [33, 40, 54] to robotic dogs [20, 36]. Furthermore, recent technological advancements in augmented and virtual reality [65], particularly in the passthrough capabilities of these devices, enable users to effortlessly overlay virtual objects and enhancements onto the physical world. This is also evident in the latest consumer devices like the Apple Vision Pro [2], Meta Quest Pro [1], and Meta Quest 3 [3] marketing their passthrough capabilities. This technology has already found diverse applications, including visual guidance [85], compensation for color blindness [50], and visual noise cancellation through vision augmentation [37]. Studies have also investigated the use cases and effectiveness of augmented reality/virtual reality for people with OVI [60, 68, 76, 94].

Considering the significance of vision-centric interaction for people with CVI, the question arises as to whether these technologies can be tailored to address their specific needs and work as assistive technologies. Could these devices leverage a combination of machine learning, image enhancement, and augmented reality to provide real-time assistance to people with CVI? For example, VBAT could be used to fade out unwanted details when locating a pair of scissors, finding friends in a crowded venue, or finding a shop on a busy street.

Hence, given the opportunities for developing VBAT specifically tailored for the needs of people with CVI, in Section 4 we dive into understanding the current research landscape in this domain.

4 LITERATURE REVIEW

Before delving further into the targeted research area (CVI and VBAT), we conducted an initial keyword search to gain insights into the research landscape pertaining to CVI. The primary objective was to obtain an overview of the prevailing trends in CVI studies. To achieve this, we executed a keyword search on Scopus using only the terms outlined in the first column ("Cerebral Visual Impairment") of Table 2a. This search was conducted on January 2, 2024, resulting in a total of 2,464 papers.

We observed an upward trajectory in the number of papers, aligning with observations made by other CVI researchers [56, 77]. Approximately 85% of the identified papers originated from disciplines closely associated with medicine, with over half exclusively within the medical domain. In stark contrast, a mere 4.9% of the papers originated from the realms of computer science or engineering. This highlights the need for the HCI and AT communities to extend research on CVI beyond the medical approach.

4.1 Scoping Review

Given this information and the potential opportunities highlighted for Vision-Based Assistive Technologies (VBAT) for people with CVI in Section 3, our subsequent aim was to gain a comprehensive understanding of the research landscape within this specific domain. To achieve this objective, we conducted a scoping review, examining the intersection of CVI with technologies for VBAT such as artificial intelligence, computer vision, image enhancement, and augmented/virtual reality. A scoping review was chosen as a method that is well-suited for examining research practices in a specific topic, identifying knowledge gaps, and categorising available evidence within a given field [67]. After identifying these studies, our overarching goal was to sift through the literature and pinpoint a more focused subset of papers that specifically delved into the assistive technologies situated at the intersection of CVI and these technologies.

Our search was structured around two key criteria: Cerebral Visual Impairment and Technologies for VBAT. The keywords, as displayed in Table 2a, were combined using "OR" within the same column and "AND" across columns during the search process. We conducted this search across four databases: Google Scholar, Scopus, Web of Science, and PubMed. The search captured papers published up to January 2, 2024.

Our search yielded 595 papers from Google Scholar, 210 from Scopus, 9 from Web Of Science, and 5 from PubMed. After consolidating and removing duplicates, a title and abstract screening process was conducted by one researcher with consultation from other members of the research team. This led to 67 papers for full-text review. Two researchers then reviewed the full texts using the criteria in Table 2b, discussing and resolving any conflicts to identify a final set of 14 papers at the intersection of CVI and technologies for VBAT.

Data Extraction: The following information was recorded for each paper:

- **Type of Study:** The primary focus of the study: Diagnosis (aiming to diagnose CVI), Understanding CVI (focused on understanding CVI), Simulation (simulating CVI conditions), Assistance (aimed at assisting users during use), or Rehabilitation (providing rehabilitation or therapy). These categories were determined by the two researchers conducting the full text review.
- **Involvement of CVI Participants:** Whether people with CVI were part of the study and, if so, whether their participation was during the design phase, evaluation phase, or both.
- **Type of Technology:** Specific type of technologies employed in the study.

Table 2: Scoping review keywords and criteria

(a) Keywords		(b) Inclusion and Exclusion Criteria	
Cerebral Visual Impairment	Technologies for VBAT	Inclusion	Exclusion
"cerebral visual impairment" "cortical visual impairment" "neurological vision impairment"	"computer vision" "artificial intelligence" "machine learning" "image enhancement" "augmented reality" "virtual reality" "mixed reality"	Papers focusing on CVI and technologies in Table 2a, irrespective of whether CVI participants are involved. Papers with CVI participants, regardless if focus is on CVI.	Theses or Non-English papers. Review papers and citations linked by references. Papers without any technologies in Table 2a.

- **Age Group and Other Details:** The participant age group: adults (over 18 years), children (below 18 years), both or unclear, and additional information, including publication venue, publication year, and authors' names.

4.2 Results

This section offers a summary of the main findings while the full dataset for all 14 papers is available in Table 3 (in the Appendix). Figure 2a shows an increase in the publication trend over time, yet making definitive statements is challenging due to the limited number of papers.

Type of Study: Figure 2d displays the distribution of study types extracted from the 14 papers. Some papers fell into multiple categories, hence the total count exceeds 14.

We found only three VBAT papers, with just one directly addressing CVI requirements. Birnbaum et al. [18] suggested that presenting high contrast and low spatial frequency visual stimuli could increase visual awareness for people with CVI. They also suggested that augmented/virtual reality headsets could modify real-time visual input to enhance visual detection. However, their proposal was not implemented.

The other two assistance studies were not specifically focused on addressing concerns related to people with CVI [55, 75]. Pitt and McCarthy [75] identified strategies for highlighting items within visual scene displays to support augmentative and alternative communication access. They identified four methods for highlighting items: contrast based on light and dark, contrast based on colour, outline highlighting, and the utilisation of scale and motion. In their future directions, they suggested the potential application of these strategies for people with CVI. Lorenzini et al. [55] conducted a study with the objective to gather insights to reduce the likelihood of device abandonment when using portable head-mounted displays for telerehabilitation. While the primary focus of their study was ocular low vision, participants with CVI were included in the study. Despite this inclusion, the study does not provide specific findings or discussions related to CVI.

Of the other studies, three focused on diagnosing CVI: Soni and Wao [82] demonstrated the effectiveness of a convolutional neural network model for identifying and gaining insights into CVI. Bennett and colleagues [8, 58] developed two novel virtual reality based visual search tasks to objectively assess higher order processing abilities in CVI. Using the same tasks, they conducted

many studies to expand the understanding of CVI. All 8 studies identified for understanding CVI in the review were from Bennett and colleagues [8, 10, 12, 13, 15, 27, 72, 99].

Involvement of CVI Participants: As depicted in Figure 2b, 10 out of the 14 papers incorporated CVI participants in the studies. However, participant involvement was primarily in the studies that were focused on understanding CVI and diagnosis. Such involvement in medical research is common practice, and in the context of these papers, participants were predominantly engaged for the purpose of validating hypotheses or effectiveness of diagnosis. None of the studies involved participants in the design or requirement gathering stages (as shown in Figure 2c).

Type of Technology: Figure 2e offers an overview of the technologies employed in the examined papers. Virtual reality emerged as the predominant technology, largely due to the contributions by Bennett and colleagues. However, focusing on the subset of the three VBAT papers, computer vision, image enhancement, and machine learning were consistently proposed or employed. However, with only three papers, it is evident that these technologies are underutilised, presenting an opportunity for future studies to address this gap.

Age Group and Other Details: In the 7 studies including both adults and children (see Figure 2b), ages ranged from 7 to 28 years old. The focus on the pediatric and young adult population is understandable given the higher CVI prevalence in children (refer to Section 2.1). However, future studies should include participants across all age groups to better understand the diverse strategies employed by people living with CVI.

5 DISCUSSION

The scoping review reveals an understudied area at the intersection of VBAT and the unique requirements of people with CVI. With CVI set to be the leading cause of visual impairment [19] and recent improvements in many of the technologies for VBAT, it is the opportune time to explore and investigate this under-researched area. Therefore, we propose to bridge this gap from two key perspectives:

Inclusion of CVI participants in ocular low vision focused assistive device research: Given the shared reliance on the visual modality and the availability of numerous assistive devices for ocular low vision [49, 100–103], evaluating their effectiveness for

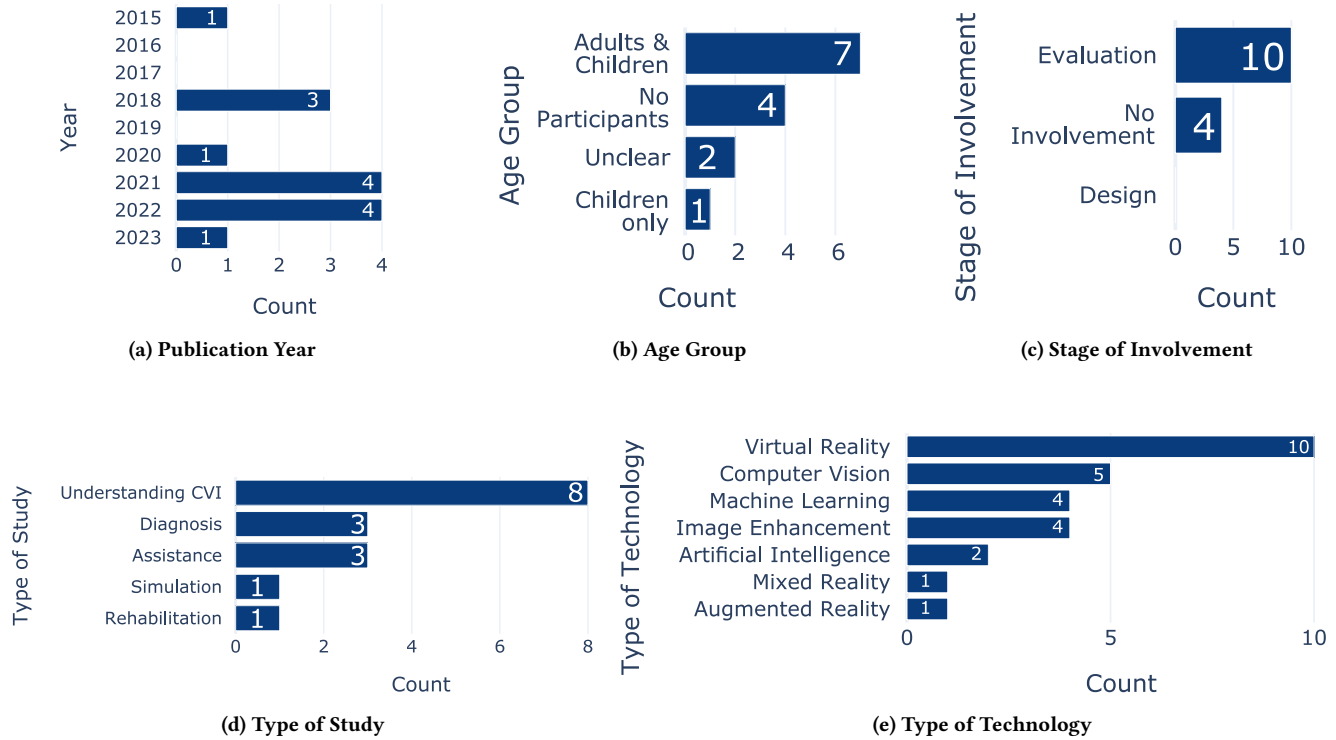


Figure 2: Summary of findings from the Scoping Review

people with CVI is essential. However, we recommend separate reporting of findings and observations for people with CVI. Moreover, given the potential permanent alterations in visual pathways due to neuroplasticity, it is essential to consider the long-term effects of using such assistive devices by people with CVI.

Co-designing devices with a focus on CVI: With the unique requirements of people with CVI such as vision-centric interaction and complexity impact on vision, it is pivotal to involve them from the requirements gathering/design to the evaluation stages. Existing research in the field of HCI has consistently demonstrated the advantages of involving end-users in the design process, ranging from superior idea generation to shorter development cycles and increased user satisfaction [11, 46, 84]. Duckett & Pratt [23] also emphasised that the visually impaired community believe that researchers could make a substantial contribution by involving them in research design and practice. Therefore, VBAT has a significant opportunity to involve people with CVI from the start when designing devices for them.

6 LIMITATIONS, FUTURE WORK, AND CONCLUSION

It is challenging to draw definitive trends from only 14 papers, however this small number clearly demonstrates the significant lack of research in this important research area, with only three papers addressing VBAT.

For future studies, we intend to adopt a human-centered approach, shifting away from a purely medical perspective. Our plan

is to engage in a co-design processes with people living with CVI, exploring methods introduced by Pitt and McCarthy [75] along with other techniques discovered through the co-design process.

This paper underscored the unique requirements of people living with CVI, highlighted opportunities for VBAT, identified a significant research gap, and proposed two approaches to address it. With recent technological advancements for VBAT and a wide-open field for research, now is the ideal time for researchers to create a substantial impact on people's lives. As CVI emerges as a prominent vision impairment, we call upon researchers in the fields of HCI and AT to recognise and address this gap.

ACKNOWLEDGMENTS

This project was funded by the Monash Data Futures Institute.

REFERENCES

- [1] 2022. Meta Connect 2022: Meta Quest Pro, More Social VR and a Look Into the Future. <https://about.fb.com/news/2022/10/meta-quest-pro-social-vr-connect-2022/>
- [2] 2023. Introducing Apple Vision Pro: Apple's first spatial computer. <https://www.apple.com/au/newsroom/2023/06/introducing-apple-vision-pro/>
- [3] 2023. Meet Meta Quest 3, Our Mixed Reality Headset. <https://about.fb.com/news/2023/09/meet-meta-quest-3-mixed-reality-headset/>
- [4] Ali Abdolrahmani, Maya Howes Gupta, Mei-Lian Vader, Ravi Kuber, and Stacy Branham. 2021. Towards More Transactional Voice Assistants: Investigating the Potential for a Multimodal Voice-Activated Indoor Navigation Assistant for Blind and Sighted Travelers. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [5] Yahya Qasim I Al-Fadhili, Paul WH Chung, Baihua Li, and Richard Bowman. 2018. 3D simulation of navigation problem of people with cerebral visual impairment. In *Advances in Computational Intelligence Systems: Contributions*

- Presented at the 17th UK Workshop on Computational Intelligence, September 6–8, 2017, Cardiff, UK. Springer, 265–275.
- [6] Susann Andersson, Eva-Karin Persson, Eva Aring, Barbro Lindquist, Gordon N Dutton, and Ann Hellström. 2006. Vision in children with hydrocephalus. *Developmental medicine and child neurology* 48, 10 (2006), 836–841.
 - [7] Adwitiya Arora, Atul Grover, Raksha Chugh, and S Sofana Reka. 2019. Real time multi object detection for blind using single shot multibox detector. *Wireless Personal Communications* 107 (2019), 651–661.
 - [8] Melissa Bambery, Camille S Corre, Daniel Kelly, Amanda Nagy, Catherine Becker, Claire Manley, Christopher Bennett, Lotfi B Merabet, and Florian Eichler. 2022. Virtual Reality Testing to Assess Cerebral Visual Impairment in X-linked Adrenoleukodystrophy (P1-5.001).
 - [9] Fereshteh S Bashiri, Eric LaRose, Jonathan C Badger, Roshan M D'Souza, Zeyun Yu, and Peggy Peissig. 2018. Object detection to assist visually impaired people: A deep neural network adventure. In *Advances in Visual Computing: 13th International Symposium, ISVC 2018, Las Vegas, NV, USA, November 19–21, 2018, Proceedings 13*. Springer, 500–510.
 - [10] Christopher Bennett, Corinna Bauer, Peter Bex, Davide Bottari, and LOTFI MERABET. 2020. Alpha Oscillatory Activity Associated with Impaired Visual Search in Cerebral Visual Impairment. (2020).
 - [11] Cynthia L. Bennett and Daniela K. Rosner. 2019. The Promise of Empathy: Design, Disability, and Knowing the "Other". In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300528>
 - [12] Christopher R Bennett, Emma S Bailin, Timothy K Gottlieb, Corinna M Bauer, Peter J Bex, and Lotfi B Merabet. 2018. Assessing visual search performance in ocular compared to cerebral visual impairment using a virtual reality simulation of human dynamic movement. In *Proceedings of the Technology, Mind, and Society*. 1–6.
 - [13] Christopher R Bennett, Emma S Bailin, Timothy K Gottlieb, Corinna M Bauer, Peter J Bex, and Lotfi B Merabet. 2018. Virtual reality based assessment of static object visual search in ocular compared to cerebral visual impairment. In *Universal Access in Human-Computer Interaction. Virtual, Augmented, and Intelligent Environments: 12th International Conference, UAHCI 2018, Held as Part of HCI International 2018, Las Vegas, NV, USA, July 15-20, 2018, Proceedings, Part II 12*. Springer, 28–38.
 - [14] Christopher R Bennett, Corinna M Bauer, Emma S Bailin, and Lotfi B Merabet. 2020. Neuroplasticity in cerebral visual impairment (CVI): assessing functional vision and the neurophysiological correlates of dorsal stream dysfunction. *Neuroscience & Biobehavioral Reviews* 108 (2020), 171–181.
 - [15] Christopher R Bennett, Corinna M Bauer, Peter J Bex, Davide Bottari, and Lotfi B Merabet. 2021. Visual search performance in cerebral visual impairment is associated with altered alpha band oscillations. *Neuropsychologia* 161 (2021), 108011.
 - [16] Christopher R Bennett, Peter J Bex, Corinna M Bauer, and Lotfi B Merabet. 2019. The assessment of visual function and functional vision. In *Seminars in pediatric neurology*, Vol. 31. Elsevier, 30–40.
 - [17] Jay S Bhatia, Nimit K Vasavat, Manali U Maniyar, Neha N Doshi, and Ruhina Karani. 2022. Audio-Based Staircase Navigation System for Visually Impaired. In *Machine Learning and Autonomous Systems: Proceedings of ICMLAS 2021*. Springer, 411–424.
 - [18] Faith A Birnbaum, Steven A Hackley, and Lenworth N Johnson. 2015. Enhancing visual performance in individuals with cortical visual impairment (homonymous hemianopia): Tapping into blindsight. *Journal of Medical Hypotheses and Ideas* 9, 2 (2015), S8–S13.
 - [19] Daniëlle GM Bosch, F Nienke Boonstra, Michèl AAP Willemsen, Frans PM Cremers, and Bert de Vries. 2014. Low vision due to cerebral visual impairment: differentiating between acquired and genetic causes. *BMC ophthalmology* 14, 1 (2014), 1–9.
 - [20] Diego Renan Bruno, Marcelo Henrique de Assis, and Fernando Santos Osório. 2019. Development of a mobile robot: Robotic guide dog for aid of visual disabilities in urban environments. In *2019 Latin American Robotics Symposium (LARS), 2019 Brazilian Symposium on Robotics (SBR) and 2019 Workshop on Robotics in Education (WRE)*. IEEE, 104–108.
 - [21] Arvind Chandna, Saeideh Ghahghaei, Susan Foster, and Ram Kumar. 2021. Higher visual function deficits in children with cerebral visual impairment and good visual acuity. *Frontiers in Human Neuroscience* 15 (2021), 711873.
 - [22] Ching-Han Chen and Ming-Fang Shiu. 2020. Smart guiding glasses with descriptive video service and spoken dialogue system for visually impaired. In *2020 IEEE International Conference on Consumer Electronics-Taiwan (ICCE-Taiwan)*. IEEE, 1–2.
 - [23] Paul S. Duckett and Rebekah Pratt. 2001. The Researched Opinions on Research: Visually impaired people and visual impairment research. *Disability & Society* 16, 6 (2001), 815–835. <https://doi.org/10.1080/09687590120083976> arXiv:<https://doi.org/10.1080/09687590120083976>
 - [24] GN Dutton. 2003. Cognitive vision, its disorders and differential diagnosis in adults and children: knowing where and what things are. *Eye* 17, 3 (2003), 289–304.
 - [25] Gordon N Dutton. 2011. Structured history taking to characterize visual dysfunction and plan optimal habilitation for children with cerebral visual impairment. *Developmental Medicine & Child Neurology* 53, 5 (2011), 390–390.
 - [26] Gordon N Dutton, A Saeed, Badsha Fahad, Richie Fraser, Gizelda McDaid, Jenny McDade, Anna Mackintosh, Talal Rane, and Katherine Spowart. 2004. Association of binocular lower visual field impairment, impaired simultaneous perception, disordered visually guided motion and inaccurate saccades in children with cerebral visual dysfunction—a retrospective observational study. *Eye* 18, 1 (2004), 27–34.
 - [27] Alessandra Federici, Christopher R Bennett, Corinna M Bauer, Claire E Manley, Emiliano Ricciardi, Davide Bottari, and Lotfi B Merabet. 2022. Altered Neural Oscillations Underlying Visuospatial Processing in Cerebral Visual Impairment (CVI). *medRxiv* (2022), 2022–10.
 - [28] Gordon N Dutton MD FRCOPHTH. 2009. 'Dorsal stream dysfunction' and 'dorsal stream dysfunction plus': a potential classification for perceptual visual impairment in the context of cerebral visual impairment? *Developmental medicine and child neurology* 51, 3 (2009), 170.
 - [29] Bhanuka Gamage, Thanh-Toan Do, Nicholas Seow Chiang Price, Arthur Lowery, and Kim Marriott. 2023. What do Blind and Low-Vision People Really Want from Assistive Smart Devices? Comparison of the Literature with a Focus Study. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*. Association for Computing Machinery, New York, NY, USA, Article 30, 21 pages. <https://doi.org/10.1145/3597638.3608955>
 - [30] William V Good, James E Jan, Susan K Burden, Ann Skoczenski, and Rowan Candy. 2001. Recent advances in cortical visual impairment. *Developmental medicine and child neurology* 43, 1 (2001), 56–60.
 - [31] William V Good, James E Jan, Luis DeSa, A James Barkovich, Myrka Groenvelde, et al. 1994. Cortical visual impairment in children. *Survey of ophthalmology* 38, 4 (1994), 351–364.
 - [32] Melvyn A Goodale. 2013. Separate visual systems for perception and action: a framework for understanding cortical visual impairment. *Developmental Medicine & Child Neurology* 55 (2013), 9–12.
 - [33] Renan Guarese, Franklin Bastidas, João Becker, Mariane Giambastiani, Yhonatan Iquiapaza, Lennon Macedo, Luciana Nedel, Anderson Maciel, Fabio Zambetta, and Ron van Schyndel. 2021. Cooking in the Dark: Exploring Spatial Audio as MR Assistive Technology for the Visually Impaired. In *IFIP Conference on Human-Computer Interaction*. Springer, 318–322.
 - [34] Deborah D Hatton, Eric Schwietz, Burt Boyer, and Paul Rychwalski. 2007. Babies Count: the national registry for children with visual impairments, birth to 3 years. *Journal of American Association for Pediatric Ophthalmology and Strabismus* 11, 4 (2007), 351–355.
 - [35] James V Haxby, Cheryl L Grady, Barry Horwitz, Leslie G Ungerleider, Mortimer Mishkin, Richard E Carson, Peter Herscovitch, Mark B Schapiro, and Stanley I Rapoport. 1991. Dissociation of object and spatial visual processing pathways in human extrastriate cortex. *Proceedings of the National Academy of Sciences* 88, 5 (1991), 1621–1625.
 - [36] Bin Hong, Zhangxi Lin, Xin Chen, Jing Hou, Shunya Lv, and Zhendong Gao. 2022. Development and application of key technologies for Guide Dog Robot: A systematic literature review. *Robotics and Autonomous Systems* (2022), 104104.
 - [37] Junlei Hong, Tobias Langlotz, Jonathan Sutton, and Holger Regenbrecht. 2023. Visual Noise Cancellation: Exploring Visual Discomfort and Opportunities for Vision Augmentations. *ACM Trans. Comput.-Hum. Interact.* (dec 2023). <https://doi.org/10.1145/3634699> Just Accepted.
 - [38] Mary Jane Houliston, Abdel H Taguri, Gordon N Dutton, Constantinos Hajivassiliou, and Daniel G Young. 1999. Evidence of cognitive visual problems in children with hydrocephalus: a structured clinical history-taking strategy. *Developmental medicine and child neurology* 41, 5 (1999), 298–306.
 - [39] Lea Hyvärinen. 1995. Considerations in evaluation and treatment of the child with low vision. *The American Journal of Occupational Therapy* 49, 9 (1995), 891–897.
 - [40] MT Islam, MA Rashid, M Ahmad, A Kuwana, and H Kobayashi. 2020. Design and Implementation of Smart Guided Glass for Visually Impaired People. *International Journal of Electrical and Computer Engineering (IJECE)* 9, 4 (2020).
 - [41] Rakesh Chandra Joshi, Saumya Yadav, and Malay Kishore Dutta. 2020. YOLO-v3 based currency detection and recognition system for visually impaired persons. In *2020 International Conference on Contemporary Computing and Applications (IC3A)*. IEEE, 280–285.
 - [42] Oliver Beren Kaul, Kersten Behrens, and Michael Rohs. 2021. Mobile Recognition and Tracking of Objects in the Environment through Augmented Reality and 3D Audio Cues for People with Visual Impairments. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–7.
 - [43] Akif Khan and Shah Khusro. 2021. An insight into smartphone-based assistive solutions for visually impaired and blind people: issues, challenges and opportunities. *Universal Access in the Information Society* 20 (2021), 265–298.
 - [44] N Kozes. 2010. Brain visual impairment in childhood: mini review. *Hippokratia* 14, 4 (2010), 249.

- [45] Barry S Kran, Linda Lawrence, D Luisa Mayer, and Gena Heidary. 2019. Cerebral/cortical visual impairment: a need to reassess current definitions of visual impairment and blindness. In *Seminars in pediatric neurology*, Vol. 31. Elsevier, 25–29.
- [46] Sari Kujala. 2003. User involvement: a review of the benefits and challenges. *Behaviour & information technology* 22, 1 (2003), 1–16.
- [47] Bineeth Kuriakose, Raju Shrestha, and Frode Eika Sandnes. 2020. Multimodal Navigation Systems for Users with Visual Impairments—A Review and Analysis. *Multimodal Technologies and Interaction* 4, 4 (2020), 73.
- [48] Fook Chang Lam, Fiona Lovett, and Gordon N Dutton. 2010. Cerebral visual impairment in children: a longitudinal case study of functional outcomes beyond the visual acuities. *Journal of Visual Impairment & Blindness* 104, 10 (2010), 625–635.
- [49] Florian Lang, Albrecht Schmidt, and Tonja Machulla. 2020. Augmented reality for people with low vision: symbolic and alphanumeric representation of information. In *Computers Helping People with Special Needs: 17th International Conference, ICCHP 2020, Lecco, Italy, September 9–11, 2020, Proceedings, Part I* 17. Springer, 146–156.
- [50] Tobias Langlotz, Jonathan Sutton, Stefanie Zollmann, Yuta Itoh, and Holger Regenbrecht. 2018. ChromaGlasses: Computational Glasses for Compensating Colour Blindness. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173964>
- [51] Chan-Su Lee, Jae-Ik Lee, and Han Eol Seo. 2021. Deep Learning Based Mobile Assistive Device for Visually Impaired People. In *2021 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*. IEEE, 1–3.
- [52] Manja Lehmann, Josephine Barnes, Gerard R Ridgway, John Wattam-Bell, Elizabeth K Warrington, Nick C Fox, and Sebastian J Crutch. 2011. Basic visual function and cortical thickness patterns in posterior cortical atrophy. *Cerebral cortex* 21, 9 (2011), 2122–2132.
- [53] Yifan Li, Kangsoo Kim, Austin Erickson, Nahal Norouzi, Jonathan Jules, Gerd Bruder, and Gregory F Welch. 2022. A Scoping Review of Assistance and Therapy with Head-Mounted Displays for People Who Are Visually Impaired. *ACM Transactions on Accessible Computing (TACCESS)* (2022).
- [54] Jyun-You Lin, Chi-Lin Chiang, Meng-Jin Wu, Chih-Chiung Yao, and Ming-Chiao Chen. 2020. Smart Glasses Application System for Visually Impaired People Based on Deep Learning. In *2020 Indo-Taiwan 2nd International Conference on Computing, Analytics and Networks (Indo-Taiwan ICAN)*. IEEE, 202–206.
- [55] Marie-Céline Lorenzini and Walter Wittich. 2021. Personalized telerehabilitation for a head-mounted low vision aid: a randomized feasibility study. *Optometry and Vision Science* 98, 6 (2021), 570.
- [56] Amanda Hall Lueck and Gordon Dutton. 2015. *Vision and the brain: Understanding cerebral visual impairment in children*. Vol. 2. AFB Press, American Foundation for the Blind Arlington, VA.
- [57] Catriona Macintyre-Béon, Hussein Ibrahim, Isobel Hay, Debbie Cockburn, Julie Calvert, Gordon N Dutton, and Richard Bowman. 2010. Dorsal stream dysfunction in children. A review and an approach to diagnosis and management. *Current Pediatric Reviews* 6, 3 (2010), 166–182.
- [58] Claire E Manley, Christopher R Bennett, and Lotfi B Merabet. 2022. Assessing Higher-Order Visual Processing in Cerebral Visual Impairment Using Naturalistic Virtual-Reality-Based Visual Search Tasks. *Children* 9, 8 (2022), 1114.
- [59] Maria BC Martin, Alejandro Santos-Lozano, Juan Martín-Hernández, Alberto López-Miguel, Miguel Maldonado, Carlos Baladrón, Corinna M Bauer, and Lotfi B Merabet. 2016. Cerebral versus ocular visual impairment: the impact on developmental neuroplasticity. *Frontiers in psychology* 7 (2016), 1958.
- [60] Sina Masnadi, Brian Williamson, Andrés N Vargas González, and Joseph J LaViola. 2020. Vriassist: An eye-tracked virtual reality low vision assistance tool. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 808–809.
- [61] Carey A Matsuba and James E Jan. 2006. Long-term outcome of children with cortical visual impairment. *Developmental medicine and child neurology* 48, 6 (2006), 508–512.
- [62] Nicola McDowell. 2020. *The CVI practice framework: an effective approach to supporting children with cerebral visual impairment (CVI): a thesis presented in partial fulfilment of the requirements for the degree of Doctor of Philosophy in Education at Massey University, Manawatu, Aotearoa New Zealand*. Ph.D. Dissertation. Massey University.
- [63] Nicola McDowell. 2022. Using digital technology to help identify children with brain based visual issues. *Pacific Journal of Technology Enhanced Learning* 4, 1 (2022), 36–37.
- [64] Nicola McDowell and Philippa Butler. 2023. Validation of the Austin Assessment: A screening tool for cerebral visual impairment related visual issues. *Plos one* 18, 11 (2023), e0293904.
- [65] Tram Thi Minh Tran, Shane Brown, Oliver Weidlich, Mark Billingham, and Callum Parker. 2023. Wearable Augmented Reality: Research Trends and Future Directions from Three Major Venues. *IEEE Transactions on Visualization and Computer Graphics* 29, 11 (2023), 4782–4793. <https://doi.org/10.1109/TVCG.2023.3320231>
- [66] Mortimer Mishkin, Leslie G Ungerleider, and Kathleen A Macko. 1983. Object vision and spatial vision: two cortical pathways. *Trends in neurosciences* 6 (1983), 414–417.
- [67] Zachary Munn, Micah DJ Peters, Cindy Stern, Catalin Tufanaru, Alexa McArthur, and Edoardo Aromataris. 2018. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC medical research methodology* 18 (2018), 1–7.
- [68] Eduardo Gross Muñoz, Ramon Fabregat, Jorge Bacca-Acosta, Néstor Duque-Méndez, and Cecilia Avila-Garzon. 2022. Augmented reality, virtual reality, and Game Technologies in ophthalmology training. *Information* 13, 5 (2022), 222.
- [69] American Conference on Pediatric Cortical Visual Impairment. 2019. What is CVI? <https://pcvis.vision/what-is-cvi/>
- [70] OpenAI. 2023. *ChatGPT by OpenAI*. <https://chat.openai.com/> 12th April 2023.
- [71] Oihana Otaegui, J Seybold, J Spiller, A Marconi, R Olmedo, and M Dubielzig. 2013. ARGUS: Assisting perSonal GUIDance System for people with visual impairment. In *Proceedings of the 26th International Technical Meeting of The Satellite Division of the Institute of Navigation (ION GNSS+ 2013)*. 2276–2283.
- [72] Zahide Pamir, Corinna M Bauer, Christopher R Bennett, Barry S Kran, and Lotfi B Merabet. 2021. Visual perception supported by verbal mediation in an individual with cerebral visual impairment (CVI). *Neuropsychologia* 160 (2021), 107982.
- [73] Karan Patel and Bhavesh Parmar. 2020. Assistive device using computer vision and image processing for visually impaired; review and current status. *Disability and Rehabilitation: Assistive Technology* (2020), 1–8.
- [74] Swetha Sara Philip and Gordon N Dutton. 2014. Identifying and characterising cerebral visual impairment in children: a review. *Clinical and Experimental Ophthalmology* 97, 3 (2014), 196–208.
- [75] Kevin M Pitt and John W McCarthy. 2023. Strategies for highlighting items within visual scene displays to support augmentative and alternative communication access for those with physical impairments. *Disability and Rehabilitation: Assistive Technology* 18, 8 (2023), 1319–1329.
- [76] Fabiana Sofia Ricci, Alain Boldini, Xinda Ma, Mahya Beheshti, Duane R Gersch, William H Seiple, John-Ross Rizzo, and Maurizio Porfiri. 2023. Virtual reality as a means to explore assistive technologies for the visually impaired. *PLoS Digital Health* 2, 6 (2023), e0000275.
- [77] Christine Roman-Lantzy. 2007. *Cortical visual impairment: An approach to assessment and intervention*. American Foundation for the Blind.
- [78] Shohista Saidkasimova, David M Bennett, Sandra Butler, and Gordon N Dutton. 2007. Cognitive visual impairment with good visual acuity in children with posterior periventricular white matter injury: a series of 7 cases. *Journal of American Association for Pediatric Ophthalmology and Strabismus* 11, 5 (2007), 426–430.
- [79] Hanna EA Sakki, Naomi J Dale, Jenefer Sargent, Teresa Perez-Roche, and Richard Bowman. 2018. Is there consensus in defining childhood cerebral visual impairment? A systematic review of terminology and definitions. *British Journal of Ophthalmology* 102, 4 (2018), 424–432.
- [80] Lisbeth Sandfeld Nielsen, Liselotte Skov, and Hanne Jensen. 2007. Visual dysfunctions and ocular disorders in children with developmental delay. I. prevalence, diagnoses and aetiology of visual impairment. *Acta Ophthalmologica Scandinavica* 85, 2 (2007), 149–156.
- [81] Walter CSS Simões, Guido S Machado, André MA Sales, Mateus M de Lucena, Nasser Jazdi, and Vicente F de Lucena Jr. 2020. A review of technologies and techniques for indoor navigation systems for the visually impaired. *Sensors* 20, 14 (2020), 3935.
- [82] Brijesh K Soni and Akhilesh A Wao. 2023. Convolutional neural network model for identifying neurological visual disorder. In *Artificial Intelligence for Neurological Disorders*. Elsevier, 61–75.
- [83] S Srijia, P Kaway, T Akshara Reddy, and M Dhanalakshmi. 2020. Raspberry Pi Based Wearable Reader For Visually Impaired People with Haptic Feedback. In *2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)*. IEEE, 257–260.
- [84] Marc Steen, Menno Manschot, and Nicole De Koning. 2011. Benefits of co-design in service design projects. *International Journal of Design* 5, 2 (2011).
- [85] Jonathan Sutton, Tobias Langlotz, Alexander Plopski, Stefanie Zollmann, Yuta Itoh, and Holger Regenbrecht. 2022. Look over there! Investigating Saliency Modulation for Visual Guidance with Augmented Reality Glasses. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 81, 15 pages. <https://doi.org/10.1145/3526113.3545633>
- [86] Suzanne H Swift, Roseanna C Davidson, and Linda J Weems. 2008. Cortical Visual Impairment in Children: Presentation Intervention, and Prognosis in Educational Settings. *Teaching exceptional children plus* 4, 5 (2008), n5.
- [87] NM Taylor, LS Jakobson, D Maurer, and TL Lewis. 2009. Differential vulnerability of global motion, global form, and biological motion processing in full-term and preterm children. *Neuropsychologia* 47, 13 (2009), 2766–2778.
- [88] Jonathan D Trobe. 2001. *The neurology of vision*. Oxford university press.
- [89] Mohammad Moeen Valipoor and Angélica de Antonio. 2022. Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic

- mapping. *Universal Access in the Information Society* (2022), 1–23.
- [90] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).
- [91] Diego Villamarín and José Manuel Menéndez. 2021. Haptic glove TV device for people with visual impairment. *Sensors* 21, 7 (2021), 2325.
- [92] Sharon Whiting, James E Jan, Peter KH Wong, Olof Flodmark, Kevin Farrell, and Andrew Q McCormick. 1985. Permanent cortical visual impairment in children. *Developmental Medicine & Child Neurology* 27, 6 (1985), 730–739.
- [93] Cathy Williams, Anna Pease, Penny Warnes, Sean Harrison, Florine Pilon, Lea Hyvarinen, Stephanie West, Jay Self, John Ferris, CVI Prevalence Study Group, et al. 2021. Cerebral visual impairment-related vision problems in primary school children: a cross-sectional survey. *Developmental Medicine & Child Neurology* 63, 6 (2021), 683–689.
- [94] Hui-Yin Wu, Aurélie Calabrèse, and Pierre Kornprobst. 2021. Towards accessible news reading design in virtual reality for low vision. *Multimedia Tools and Applications* 80, 18 (2021), 27259–27278.
- [95] Jiayang Wu, Wensheng Gan, Zefeng Chen, Shicheng Wan, and Philip S Yu. 2023. Multimodal large language models: A survey. *arXiv preprint arXiv:2311.13165* (2023).
- [96] Shamima Yasmin. 2020. Haptic Selfies: Bold and Beautiful Living for the Blind and Visually Impaired. In *Proceedings of the 26th ACM Symposium on Virtual Reality Software and Technology*. 1–3.
- [97] Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. 2023. A Survey on Multimodal Large Language Models. *arXiv preprint arXiv:2306.13549* (2023).
- [98] Xiaojun Zhang, Sachin Kedar, Michael J Lynn, Nancy J Newman, and Valérie Biousse. 2006. Homonymous hemianopia in stroke. *Journal of Neuro-ophthalmology* 26, 3 (2006), 180–183.
- [99] Xin Zhang, Claire E Manley, Serena Micheletti, Isidora Tesic, Christopher R Bennett, Elisa M Fazzi, and Lotfi B Merabet. 2022. Assessing visuospatial processing in cerebral visual impairment using a novel and naturalistic static visual search task. *Research in Developmental Disabilities* 131 (2022), 104364.
- [100] Yuhang Zhao, Elizabeth Kupferstein, Brenda Veronica Castro, Steven Feiner, and Shiri Azenkot. 2019. Designing AR visualizations to facilitate stair navigation for people with low vision. In *Proceedings of the 32nd annual ACM symposium on user interface software and technology*. 387–402.
- [101] Yuhang Zhao, Elizabeth Kupferstein, Hathaitorn Rojnirun, Leah Findlater, and Shiri Azenkot. 2020. The effectiveness of visual and audio wayfinding guidance on smartglasses for people with low vision. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–14.
- [102] Yuhang Zhao, Sarit Szpiro, and Shiri Azenkot. 2015. Foresee: A customizable head-mounted vision enhancement system for people with low vision. In *Proceedings of the 17th international ACM SIGACCESS conference on computers & accessibility*. 239–249.
- [103] Yuhang Zhao, Sarit Szpiro, Jonathan Knighten, and Shiri Azenkot. 2016. CueSee: exploring visual cues for people with low vision to facilitate a visual search task. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. 73–84.

A REVIEW PAPERS

Table 3: List of 14 papers from the scoping review with the extracted data: Paper, Year, Age Group (A - Adults only, C - Children only, B - Both, ? - Unclear, N - None), Age Range ([Youngest] - [Eldest], U - Unclear, N - None), Type of Study (D - Diagnosis, U - Understanding of CVI, S - Simulation, A - Assistance, R - Rehabilitation), Involvement (D - Design Stage, E - Evaluation Stage, B - Both, N - None), Type of Technology

Study	Year	Age Group	Age Range	Type of Study	Involvement	Type of Technology
[18]	2015	N	N	A	N	Computer Vision, Image Enhancement, Augmented Reality, Mixed Reality
[13]	2018	N	14-28	U	E	Virtual Reality
[5]	2018	N	N	S	N	Computer Vision, Artificial Intelligence, Machine Learning, Virtual Reality
[12]	2018	B	14-28	U	E	Virtual Reality
[10]	2020	B	14-25	U	E	Virtual Reality
[55]	2021	?	?	A, R	E	Computer Vision, Image Enhancement, Machine Learning
[72]	2021	C	17	U	E	Virtual Reality
[75]	2021	N	N	A	N	Computer Vision, Image Enhancement, Machine Learning
[15]	2021	B	15-22	U	E	Virtual Reality
[27]	2022	B	14-21	U	E	Virtual Reality
[8]	2022	?	?	U, D	E	Virtual Reality
[99]	2022	B	7-20	U	E	Virtual Reality
[58]	2022	B	11-20	D	E	Virtual Reality
[82]	2023	N	N	D	N	Computer Vision, Artificial Intelligence, Image Enhancement, Machine Learning